

IEDA 1250 Lecture 3: Integer Programming

Instructor: Bo YANG

Summer, 2026



Some materials are from Gurobi tutorial

Today's Plan

- Motivation
- Branch and bound
- Modeling logical constraints/Traveling salesman problem (TSP)
- Solving IPs with Excel

Motivation

- Consulting company's open positions: Tester, Java-Developer, and Architect
- Three top resources: Carlos, Joe, and Monika
- Results of competency tests: matching scores
- Assumption: Only one resource can be assigned to a job, and at most one job can be assigned to a resource.

Matching Scores	Tester	Java-Developer	Architect
Carlos	53%	27%	13%
Joe	80%	47%	67%
Monika	53%	73%	47%

Motivation

- How will you assign these three people to maximize the total score? How many solutions can you find?

Matching Scores		Tester	Java-Developer	Architect
Carlos		53%	27%	13%
Joe		80%	47%	67%
Monika		53%	73%	47%

Number	Assignment	Score
1	Tester - Carlos (53%) Java - Joe (47%) Archt.- Monika (47%)	147%
2	Tester- Joe (80%) Java - Carlos (27%) Archt.- Monika (47%)	154%
3	Tester - Monika (53%) Java - Joe (47%) Archt.- Carlos (13%)	113%
4	Tester - Carlos (53%) Java - Monika (73%) Archt.- Joe (67%)	193%
5	Tester - Joe (80%) Java - Monika (73%) Archt.- Carlos (13%)	166%
6	Tester - Monika (53%) Java - Carlos (27%) Archt.- Joe (67%)	147%

- Answer: $3! = 3 * 2 * 1 = 6$
- We can solve this problem by enumeration (brute force)

Motivation

- Now suppose that the consulting company wins a major government contract that requires to assign 100 people to 100 jobs
- How many feasible solutions does this problem have?
- Answer: $100! = 100 * 99 * \dots * 1 = 9.33^{157}$
- Take thousands of years for supercomputers to find the optimal solution

Motivation

- What if we **ignore** the integer requirement and solve the problem as an LP?

Matching Scores	Tester	Java-Developer	Architect
Carlos 	53%	27%	13%
Joe 	80%	47%	67%
Monika 	53%	73%	47%

- Decision variables

Decision Variables	Tester = 1	Java-Developer = 2	Architect = 3
Carlos = 1 	$x_{1,1}$	$x_{1,2}$	$x_{1,3}$
Joe = 2 	$x_{2,1}$	$x_{2,2}$	$x_{2,3}$
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Motivation

- Suppose $x_{i,j}$ is our decision variable, i.e., $x_{i,j}$ is between 0 and 1, for $i=1,2,3$, and $j=1,2,3$

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- Constraint (Tester = 1): $x_{1,1} + x_{2,1} + x_{3,1} = 1$

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Motivation

■ Objective function

- Job 1 (Tester) matching score: $(53x_{1,1} + 80x_{2,1} + 53x_{3,1})$

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Motivation

■ Objective function

- Job 1 (Tester) matching score: $(53x_{1,1} + 80x_{2,1} + 53x_{3,1})$
- Job 2 (Java Developer) matching score: $(27x_{1,2} + 47x_{2,2} + 73x_{3,2})$

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- Job 2 (Java Developer) matching score: $(27x_{1,2} + 47x_{2,2} + 73x_{3,2})$
- Job 3 (Architect): $(13x_{1,3} + 67x_{2,3} + 47x_{3,3})$

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- Job 3 (Architect): $(13x_{1,3} + 67x_{2,3} + 47x_{3,3})$
- **Max $(53x_{1,1} + 80x_{2,1} + 53x_{3,1}) + (27x_{1,2} + 47x_{2,2} + 73x_{3,2}) + (13x_{1,3} + 67x_{2,3} + 47x_{3,3})$**

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- Optimal solution to LP: $x_{1,1}=1, x_{2,3}=1, x_{3,2}=1$
- The above solution is optimal even if we consider the integer requirement, i.e., we restrict $x_{i,j}$ to be either 0 and 1. **Why? Is this always true?**

■ Objective function

- Job 1 (Tester) matching score: $(53x_{1,1} + 80x_{2,1} + 53x_{3,1})$
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- Optimal solution: $x_{1,1}=1, x_{2,3}=1, x_{3,2}=1$
- For IPs, if the constraint coefficients satisfy certain condition and the RHS is integral, then the optimal solution is integral

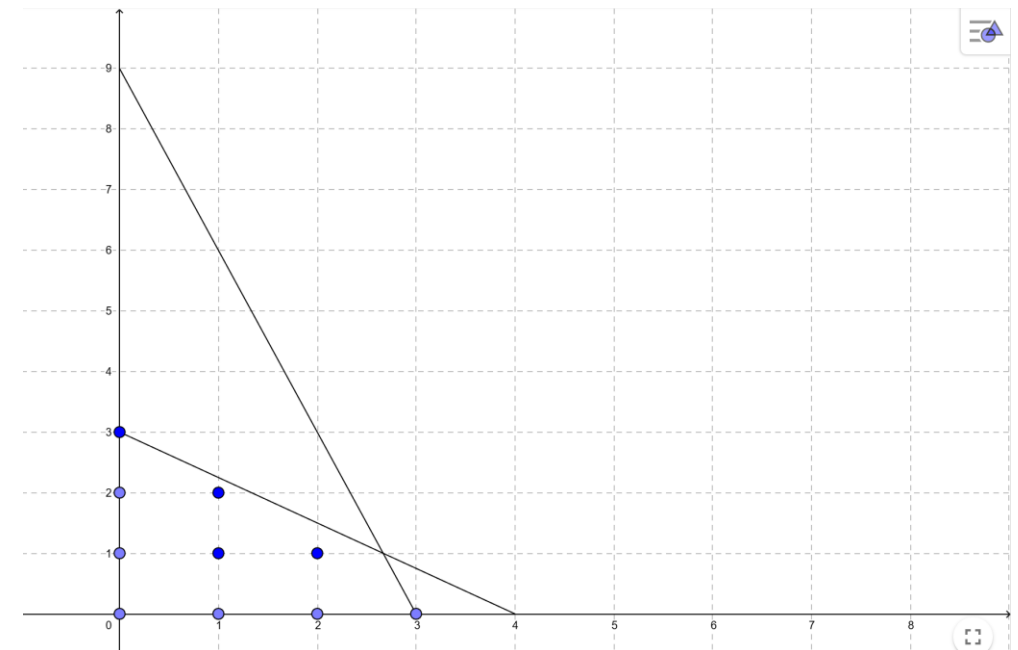
Another Example

- A large factory makes tables and chairs. Each table returns a profit of \$200 and each chair a profit of \$100. Each table takes 1.5 unit of metal and 3 units of wood, and each chair takes 2 units of metal and 1 unit of wood. The factory has 6 units of metal and 9 units of wood. How many tables and chairs should the factory make to maximize profit?

- **LP Relaxation:**

$$\begin{aligned} \max_{x_1, x_2} \quad & 200x_1 + 100x_2 \\ \text{s.t.} \quad & 1.5x_1 + 2x_2 \leq 6 \\ & 3x_1 + x_2 \leq 9 \\ & x_1, x_2 \geq 0 \end{aligned}$$

- Optimal solution to the LP: $x_1=2.667, x_2=1$
- Optimal integer solution: $x_1=3, x_2=0$



Integer (Linear) Programming

- An integer (linear) program maximizes a linear objective function

$$\max_{x_1, \dots, x_n} f(x_1, x_2, \dots, x_n) = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

where $c_1, c_2, \dots, c_n$ are constants

- It is subject to linear constraints

$$G(x_1, x_2, \dots, x_n) \leq b \text{ or } G(x_1, x_2, \dots, x_n) \geq b$$

and the integer requirement

$$x_1, x_2, \dots, x_n \in \mathbb{Z}$$

where $G(x_1, x_2, \dots, x_n)$ is a linear function

- If variables can be both integer and non-integer, it is called **mixed integer (linear) programming**
- If variables are either 0 or 1, it is called **binary integer programming**

Branch and Bound

- Solving large scale IPs is **difficult** in general
 - In the worse case, we can not do better than enumeration
 - In practice, it is possible to solve the problem fast
 - In most cases, finding a near optimal solution is good enough
- Branch and bound method repeats the following steps
 - Choose an LP relaxation: Ignore the integer requirement of the IP
 - Solve the LP relaxation
 - If the optimal solution is integral, stop
otherwise, pick a non-integer variable, add constraints to the LP relaxation, and go to the 2nd step

Branch and Bound

$$\begin{aligned} \min_{x_1, x_2, x_3} \quad & 2x_1 + x_2 - 2x_3 \\ \text{s.t.} \quad & 0.7x_1 + 0.5x_2 + x_3 \geq 1.8 \\ & x_1, x_2, x_3 \in \{0, 1\} \end{aligned}$$

Branch and Bound

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Branch and Bound

LP Relaxation

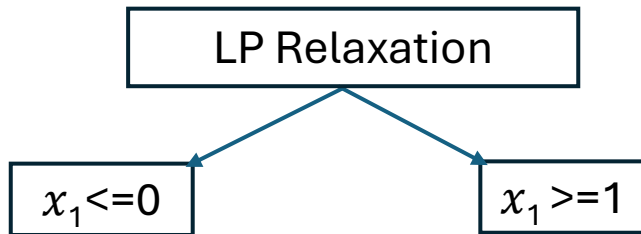
Solutions

OBJ = -0.143, (0.43, 1.0, 1.0)

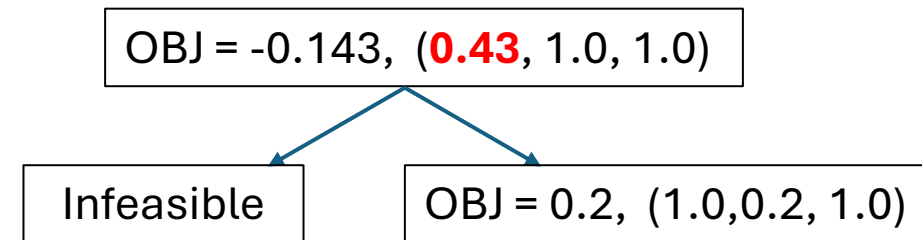
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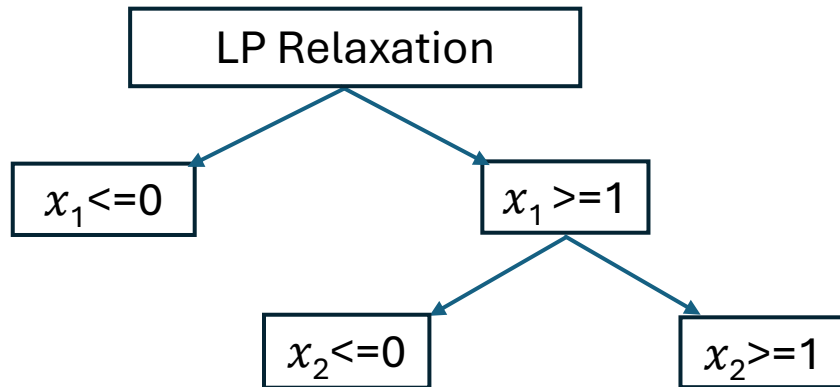
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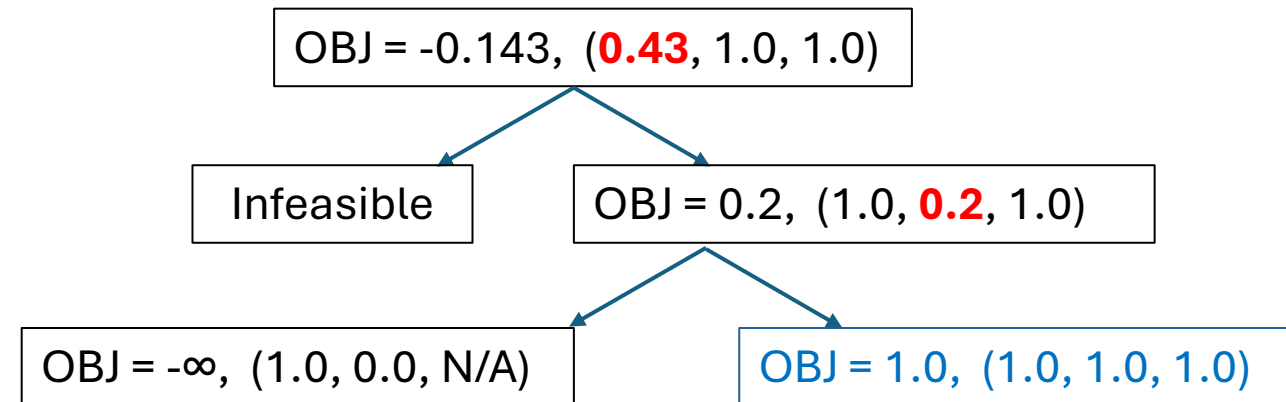
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Branch and Bound



Solutions



Branch and Bound

$$\begin{aligned} \max_{x_1, x_2} \quad & 40x_1 + 90x_2 \\ \text{s.t.} \quad & 9x_1 + 7x_2 \leq 56 \\ & 7x_1 + 20x_2 \leq 70 \\ & x_1, x_2 \in \mathbb{Z}^+ \end{aligned}$$

Branch and Bound

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Branch and Bound

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LP Relaxation

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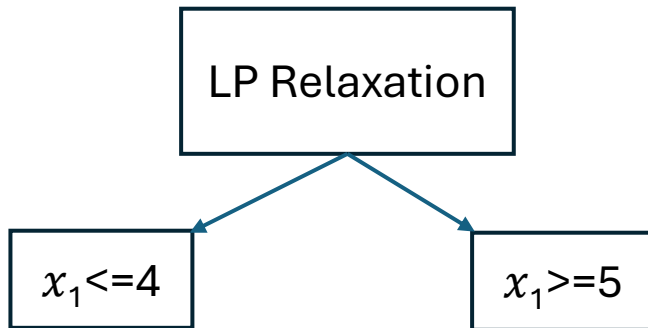
Solutions

OBJ = 356, (**4.81**, 1.82)

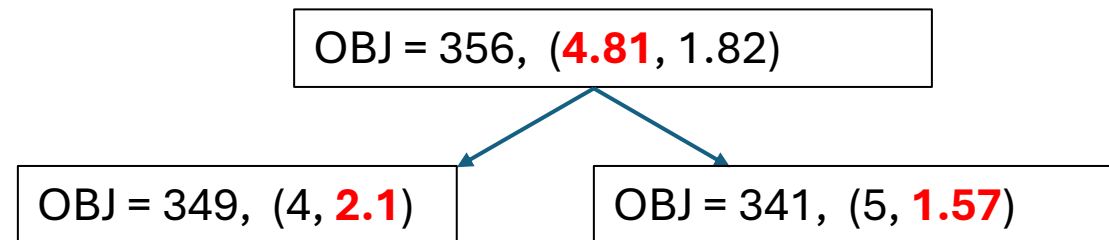
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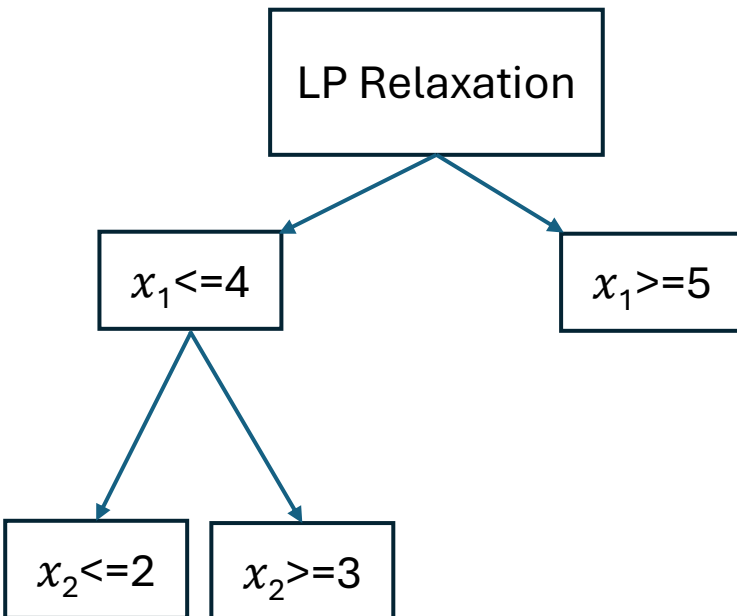
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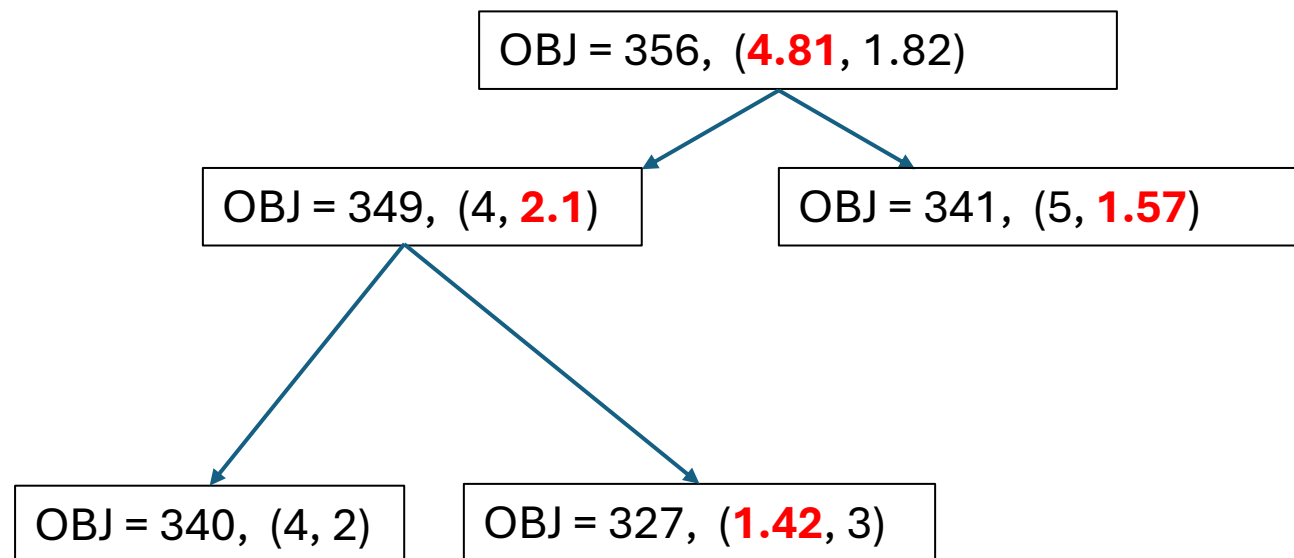
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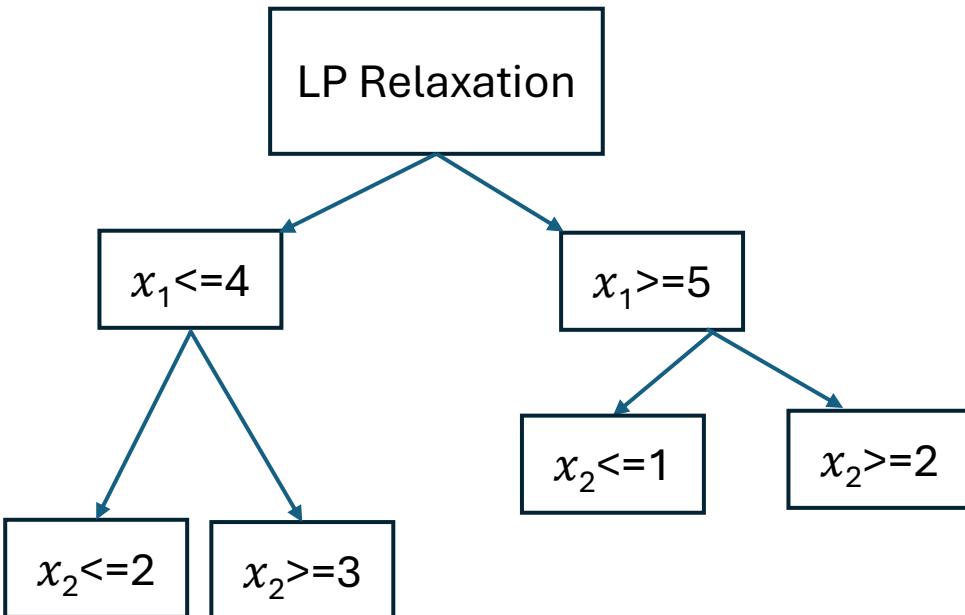
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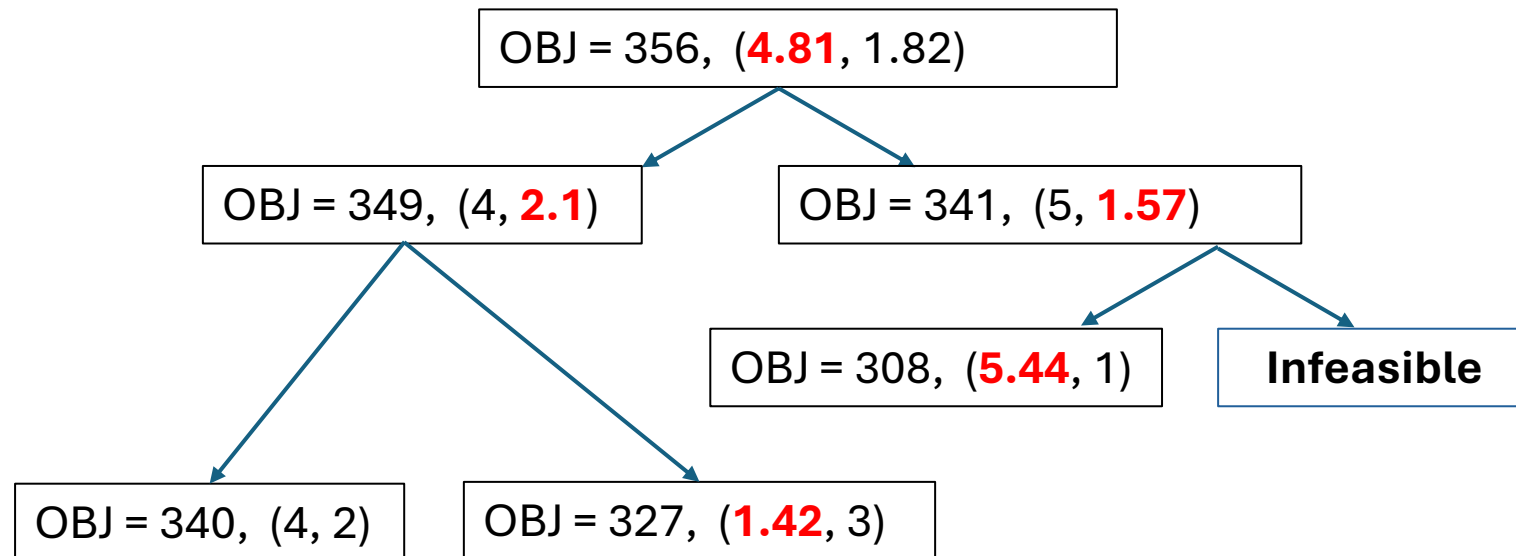
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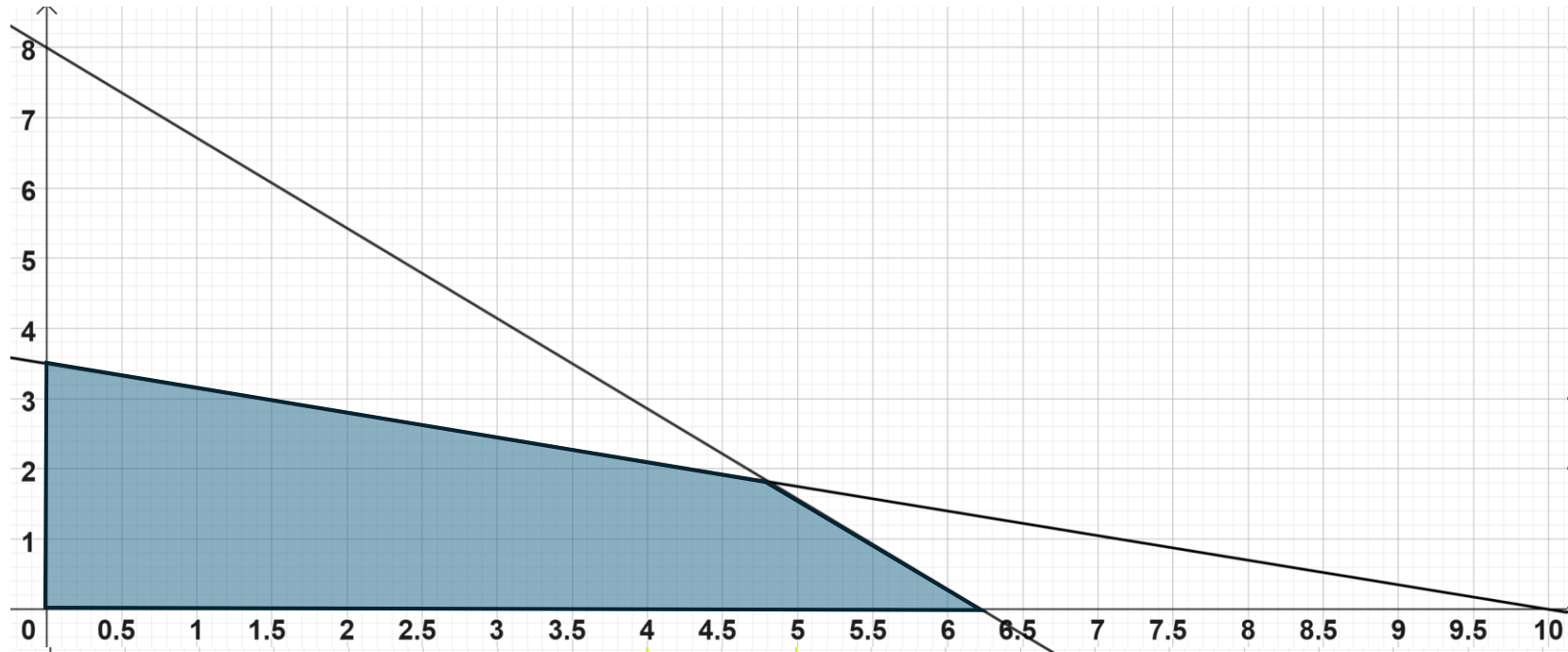


Revisit the B&B example

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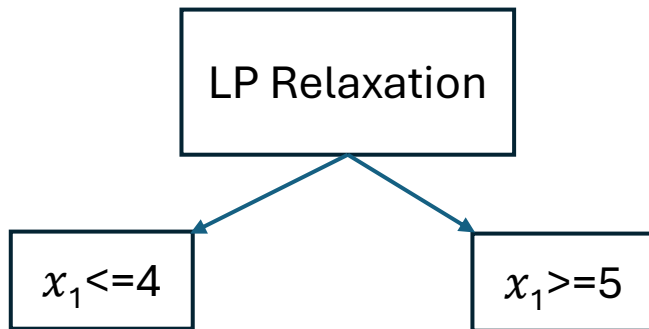
LP Relaxation



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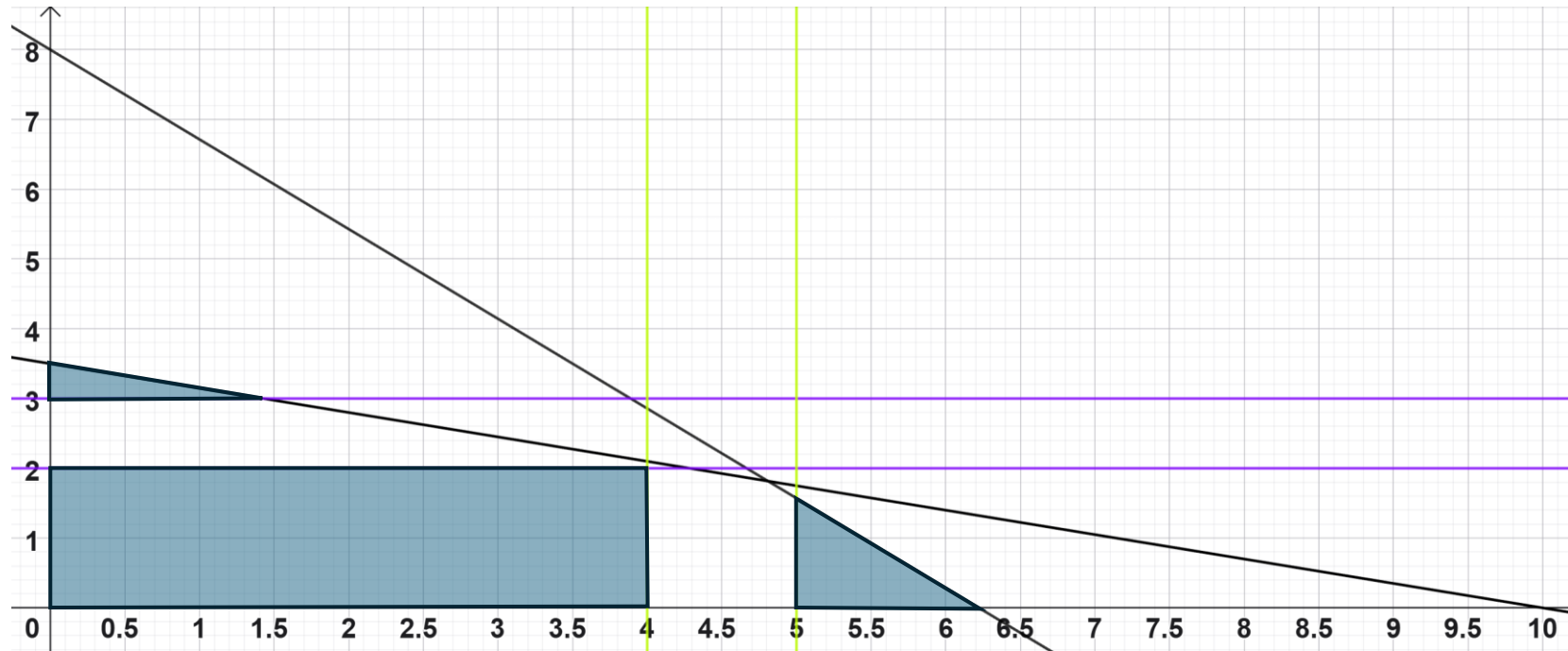
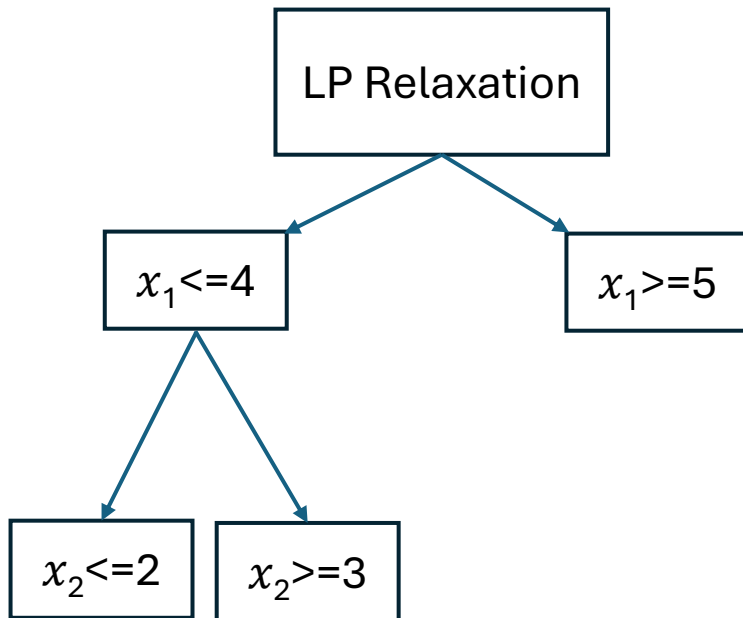
Branch and Bound



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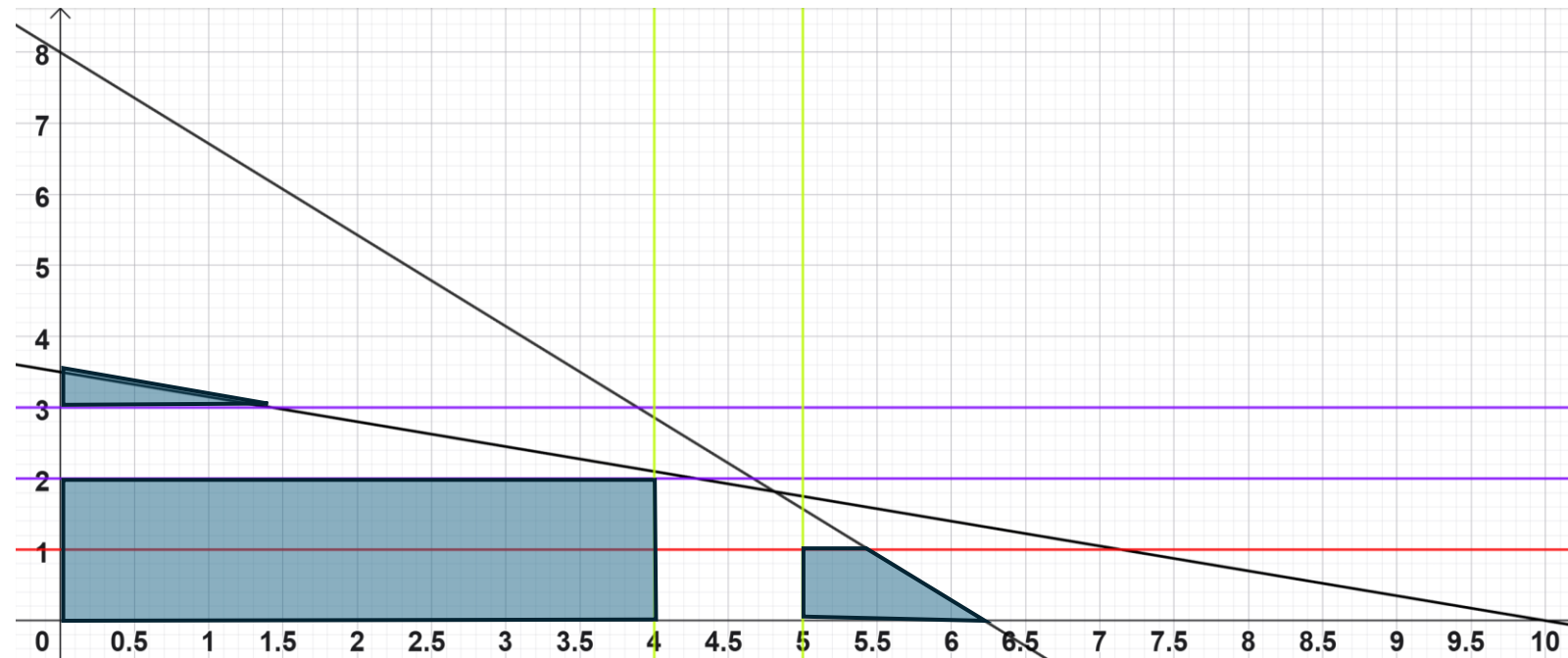
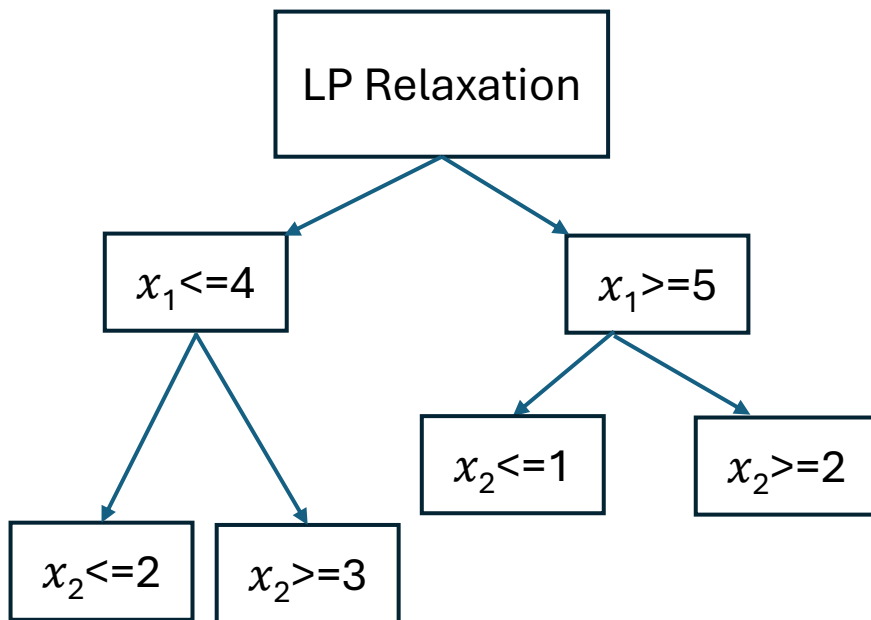
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Branch and Bound



Branch and Bound

■ Remarks

- The optimal objective function value of the LP relaxation will be better than that of the IP
- Once we add constraints, the objective function value becomes worse

■ Procedures

- Solve an LP relaxation of the IP.
 - If the solution is integer, then it is done; otherwise, create two new subproblems by branching on a fractional variable.
 - A subproblem is not active when any of the following occurs:
 - Its solution is integral
 - It is infeasible or unbounded
 - Its optimal objective function value is worse than that of the obtained integer solution
- Choose an active node and branch on a fractional variable. Repeat until there are no active subproblems.

More on IP Models

- Modeling logical constraints with IP
- Traveling salesman problem (TSP)

More on IP Models

■ Modeling logical constraints with IP

- Example 1: Suppose a company wants to build **three** warehouses at Hong Kong. There are **seven** potential locations, denoted as $A_1, \dots, A_7$. This investment needs to satisfy the following conditions:

- The company can build at most two warehouses among A_1, A_2 , and A_3
- There is at least a warehouse at A_4 and A_5
- There is at least a warehouse at A_6 and A_7
- The total budget can not exceed B
- For $i \in \{1, \dots, 7\}$, if the company chooses to build a warehouse at A_i , the corresponding investment amount is b_i and the estimated profit is c_i

$$\begin{aligned} \max \quad & \sum_{i=1}^7 c_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^7 b_i x_i \leq B \\ & x_1 + x_2 + x_3 \leq 2 \\ & x_4 + x_5 \geq 1 \\ & x_6 + x_7 \geq 1 \\ & x_i \in \{0, 1\} \end{aligned}$$

■ Modeling logical constraints with IP

- Example 2: Suppose a company wants to transport two kinds of products A and B from Hong Kong to New York City. The associated information is summarized in the following table:

Product	Volume		Weight (100 kg)	Profit
	By train (m^3)	By ship (m^3)		
A	5	7	2	20
B	4	3	5	10
Capacity	24	45	13	

- Suppose also that the company can only choose one type of transportation. How many products should the company transport? What is the associated IP?

$$\begin{aligned}
 &\max_{x_1, x_2, y} \quad 20x_1 + 10x_2 \\
 &s.t. \quad 5x_1 + 4x_2 \leq 24 + yM \\
 &\quad \quad 7x_1 + 3x_2 \leq 45 + (1 - y)M \\
 &\quad \quad 2x_1 + 5x_2 \leq 13 \\
 &\quad \quad x_1, x_2 \in \mathbb{Z}^+ \\
 &\quad \quad y \in \{0, 1\}
 \end{aligned}$$

■ Modeling logical constraints with IP

- Example 3: Suppose a company wants to maximize the profit of its production. Suppose the capacity of an equipment is B , the fixed and marginal costs are K and c , respectively. That is, if the company produces x units, then the total production cost is given by

$$\text{Production cost} = \begin{cases} K + cx, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \end{cases}$$

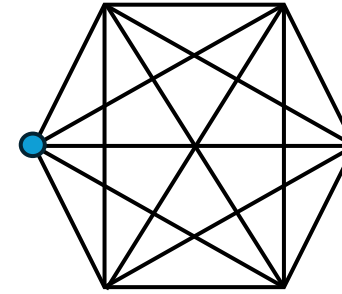
- Suppose the company sells each unit at a marginal profit p ($>c$). What is the associated IP?

$$\max_{x,y} \quad px - (Ky + cx)$$

$$s.t. \quad x \leq By$$

$$x \geq 0$$

$$y \in \{0, 1\}$$

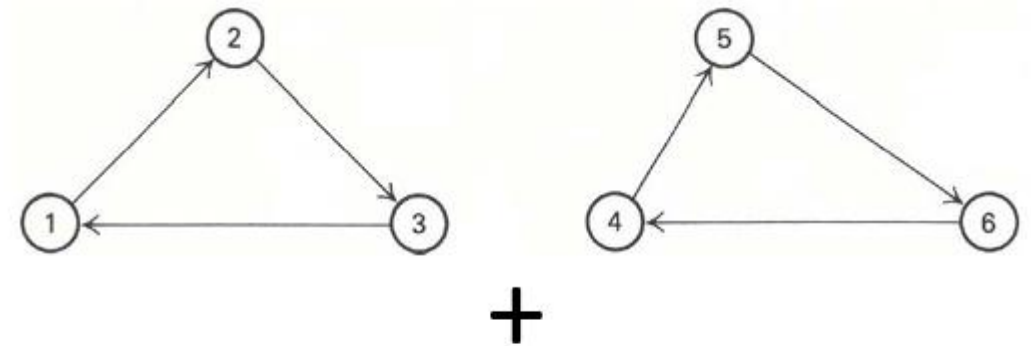


■ Traveling salesman problem (TSP)

- Suppose the traveling salesman wants to visit 6 cities.
- Starting from his home, the salesperson wishes to visit each of 5 other cities and return home at minimal cost. The following conditions need to be satisfied
 - He must visit each city exactly once and it costs $c_{i,j}$ to travel from city i to city j . For example, $c_{1,2}$ represents the cost of traveling from city 1 to city 2.
- What route should he select?

$$\begin{aligned}
 \min \quad & \sum_{i=1}^6 \sum_{j=1}^6 c_{i,j} x_{i,j} \\
 \text{s.t.} \quad & \sum_{j=1}^6 x_{i,j} = 1, \quad (i = 1, 2, \dots, 6) \\
 & \sum_{i=1}^6 x_{i,j} = 1, \quad (j = 1, 2, \dots, 6) \\
 & x_{i,j} \in \{0, 1\}
 \end{aligned}$$

We need to eliminate subtours!
Unfortunately, for n cities, there are $2^n - 2$ constraints



$$x_{14} + x_{15} + x_{16} + x_{24} + x_{25} + x_{26} + x_{34} + x_{35} + x_{36} \geq 1$$

Exercise: Capital Budgeting Problem

- Stockco is considering 5 investments. The investments will last 4 years. Each investment requires a certain cash outflows from year 1 to 4, which is shown by the following table. In year 5, Investment 1 will yield a net present value (NPV) of \$10 million; investment 2, an NPV of \$18 million; investment 3, an NPV of \$22 million; investment 4, an NPV of \$25 million; and investment 5, an NPV of 12 million. There are budget constraints, i.e., the amount of money available, in each year. In addition, the following constraints must be satisfied:
 - Unused fund in one year can be carried out to the following year
 - Investment 2 can not be selected unless Investment 1 is also selected
 - Investment 4 and 5 can not be selected together
 - At least one of Investment 3, 4, and 5 must be selected.

	P1	P2	P3	P4	P5	Budget
Year 1	5	11	13	16	9	32
Year 2	2	8	7	7	6	30
Year 3	2	6	6	6	1	15
Year 4	0	3	4	6	-1	11
NPV	10	18	22	25	12	